FIRM: FHIR-Integrated Interviewing with Replaceable Modules for LLM-Assisted Pre-Consultation

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Abstract

Purpose: Large language model (LLM) assistants for pre-consultation must behave predictably and exchange data with electronic health records (EHRs). We present ELVINA (EHR-Linked Virtual Interviewer with Natural-language Assistance), a pre-consultation system built on FIRM (FHIR-Integrated Interviewing with Replaceable Modules), in which questionnaire rules, not the LLM, select questions and results are written back as Taiwan Core (TW Core) FHIR resources, and we verify it against explicit requirements.

Methods: Questionnaires are encoded as FHIR Questionnaire. LLMs only extract evidence-grounded chief-complaint findings, classify EHR history into

questionnaire options, and draft reports; one transaction writes QuestionnaireResponse, Observation, Composition, and Provenance. Ten requirements (R0–R9) were checked with synthetic Synthea patients and 30 author-drafted scenarios answered by an LLM-simulated patient: interoperability, end-to-end runs, configuration-only model replacement, and local load. Clinician confirmation was simulated.

Results: Requirements R0, R1, R2, R3, R4, R6 passed and R5 failed. Question selection (R0) was Pass, and structural/profile checks of 1,090 written resources (R1, without terminology resolution) were Pass. After EHR-history classification was added, written-FHIR and runtime question paths agreed in 100 of 100 comparable sessions (R3, Pass). Replacing the commercial model by configuration alone was Pass; a local open-weight model was Fail. Red-flag rule coverage (R5) was Fail; candidate rules await physician approval. National-scale capacity was Not verifiable and exchange-hub integration Blocked.

Conclusions: A rule-governed pre-consultation system with FHIR write-back can be verified requirement by requirement. These provisional, synthetic software-verification results do not establish clinical safety, diagnostic performance, or deployment readiness.

Keywords: FHIR, Pre-consultation, Conversational AI, Clinical Documentation

1 Introduction

Emergency department (ED) crowding is a persistent problem worldwide, prolonging waits, delaying treatment, and worsening patient outcomes (Morley et al. 2018). In Taiwan, about one-third of the six million annual ED visits are non-emergent (Department of Statistics, Ministry of Health and Welfare, Taiwan 2024; Huang et al. 2019) [**TODO_AUTHOR: MOHW 2023 ED claims are 6.82 million (2022: 6.13 million); revise the figure or cite the 2022 data**], yet each of these patients still requires a careful history, because a small proportion of apparently minor symptoms conceal live-threatening [**TODO_AUTHOR: life-threatening?**] conditions. Taking and documenting this history consumes much of the physician's time. A study showed emergency physicians spent nearly half of their time on data and order entry, while less than one-third was spent on direct patient care (Hill et al. 2013), which has also been associated with physician burnout (Ghaseminejad et al. 2025). Pre-consultation, in which a structured history is collected while patients wait, could therefore relieve this burden, shifting the clinician's role from manual data entry to the verification of pre-extracted information (Eryılmaz et al. 2026).

Much of the patient information is already recorded elsewhere. Allergies, current medications, chronic conditions, and surgical history inform both treatment and risk stratification (Six et al. 2008), yet asking every patient for them lengthens the interview, and repeated questionnaires can frustrate respondents and produce conflicting information across records (Thayer et al. 2025) [**TODO_AUTHOR: Thayer et al. describe parent smoking-cessation decision support, not questionnaire burden; replace the citation**]. Fast Healthcare Interoperability Resources (FHIR)

offers a means to retrieve such data instead. Substitutable Medical Applications, Reusable Technologies (SMART) on FHIR allows applications to read patient context, including allergies, medications, and conditions, from an electronic health record (EHR) through a standard interface (Mandel et al. 2016). In Taiwan, the Ministry of Health and Welfare (MOHW) has published the Taiwan Core Implementation Guide (TW Core IG), a national FHIR core implementation guide, and allows consented electronic medical records from other facilities to be retrieved with FHIR-based methods (Ministry of Health and Welfare, Taiwan n.d.) [**TODO_AUTHOR: the TW Core portal does not describe consented cross-facility retrieval; cite the EMR exchange IG or MOHW policy**], making prior records available when to any visiting facility [**TODO_AUTHOR: grammar; e.g. "available to any visiting facility"**]. Retrieving what is already known and asking only for what is missing could thus shorten the interview and reduce documentation, while keeping critical history from being omitted.

Computerized history-taking has been increasingly explored to support this process. A previous study found self-reported tablet histories from 500 ED patients with acute chest pain yielded enough information to calculate a risk score in 70.4% of patients, but only 29.6% completed the full interview, with older adults showing lower completion rates (Brandberg et al. 2021). A systematic review of 18 history-taking chatbots [**TODO_AUTHOR: the review included 18 studies, not 18 chatbots**] similarly reported improved data collection and patient satisfaction, while many systems relied on rule-based scripts or decision trees, none used large language models (LLMs), and effects on clinical workflow remained unclear (Hindelang et al. 2024). Collectively, these studies show that structured history-taking is feasible, but fixed questionnaires are limited in completion by their length, highlighting the need for interviews that adapt to the patient and to what the record already contains.

LLMs have since been applied to conduct diagnostic conversations directly with patients. In simulated consultations, the Articulate Medical Intelligence Explorer (AMIE) matched or exceeded primary care physicians in history-taking and diagnostic accuracy (Tu et al. 2025), and in a prospective study of 98 primary care patients its pre-visit differential diagnosis included the final diagnosis in 90% of cases, although the system had no access to the EHR (Brodeur et al. 2026) [**TODO_AUTHOR: 98 patients is from the Lancet article, 90% from the arXiv preprint (100 patients); confirm 90% in the Lancet full text or cite the preprint**]. Conversely, MIRA, an autonomous agent, retrieved patient data and placed orders through FHIR, but it served the clinician rather than the patient and was evaluated with LLM-simulated patients (Ferber et al. 2026). All in all, patient-facing LLM interviews have so far operated apart from the EHR, while EHR-integrated agents serve clinicians, leaving the link between what the patient reports and what the record already holds largely unaddressed.

Despite their promising diagnostic performance, these systems rely on free-text generation for decisions in which errors carry direct clinical consequences, including hallucinations, omissions, and negation errors (Asgari et al. 2025) [**TODO_AUTHOR: negation errors not in the Asgari abstract; check the**

full text]. In a benchmark of 16 LLM configurations on real ED triage data, the models were generally overconfident, and gave inconsistent answers to repeated queries (Benning et al. 2026). Moreover, LLMs used to convert clinical text into FHIR resources (Li et al. 2024) remain uneven on real documents, with case-level correctness on physician-reviewed discharge letters ranging from 93.0% for medications to 61.3% for procedures (Eryılmaz et al. 2026). This highlights the need for designs in which question selection and red-flag screening follow explicit, deterministic rules that behave identically for the same input, while model output is limited to content that can be traced to the patient's own words or confirmed by a clinician before it is written back to the record.

Here we present ELVINA (EHR-Linked Virtual Interviewer with Natural-language Assistance), a modular, FHIR-integrated framework [**TODO_AUTHOR: title, abstract and Methods call FIRM the framework and ELVINA its reference implementation; unify naming and the spelling preconsultation/preconsultation**] for LLM-assisted preconsultation that links the patient interview to the clinical record. ELVINA retrieves known history from the EHR and asks only for what is missing through predefined complaint-specific questionnaires, follows explicit rules for question selection and red-flag screening, restricts the LLM to traceable extraction and clinician-reviewed charting, and writes the confirmed interview back to the record in standard FHIR resources. As a first step toward clinical use, we verified whether the framework meets these design requirements using synthetic patients and scripted scenarios, establishing its technical feasibility [**TODO_AUTHOR: after the 2026-10-15 reruns R5 (CP-09, AP-10 rule gaps) and R7 for local Gemma still fail, and R8/R9 are not verifiable/blocked; soften this claim or resolve them**].

2 Methods

2.1 Scope and implemented architecture

We verified ELVINA, FIRM's reference implementation, using synthetic data. Outcomes concern interchange, interview-to-write-back completion, model replacement, and local concurrency; clinical accuracy and note quality were outside scope.

We stated ten software requirements (R0–R9; Table 2) and tested them in five verification activities (E1–E5): E1, conformance and write/read-back of generated FHIR resources (R1, R2); E2, end-to-end sessions in which an LLM-simulated patient answers 30 scripted scenarios (R0, R3–R6); E3, E2 repeated after replacing the default model by configuration (R7); E4, local concurrent-load replay (R8); and E5, integration with the MOHW exchange hub, which could not be run (R9).

ELVINA combines Vue patient and clinician interfaces, a FastAPI backend, local retrieval, and a HAPI FHIR R4 sandbox (Fig. 1) (Vue.js 2026; Ramírez 2026; HAPI FHIR Project 2026; HL7 International 2019b). Invitations yield scoped session cookies; the clinician launcher binds patient and encounter identifiers. The separate SMART surface uses another authorization flow (Mandel et al. 2016; HL7 International 2024). Verification exercised text HTTP APIs, using isolated databases and an explicit loopback development authorization bypass.

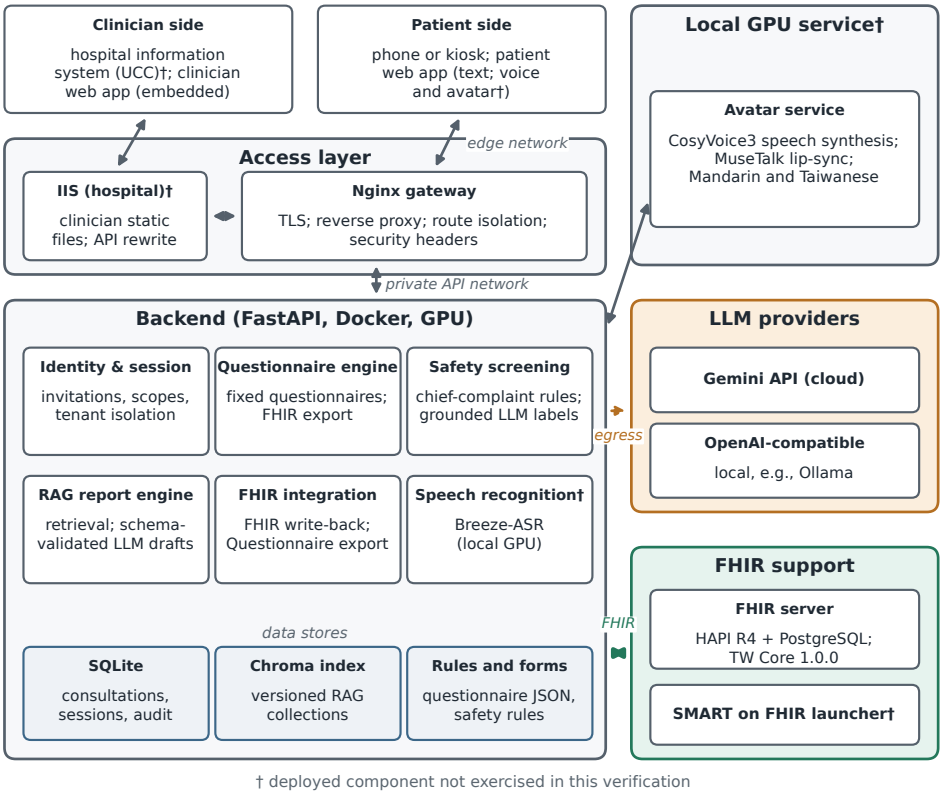


Fig. 1 ELVINA system architecture. Clinician traffic passes through the hospital IIS and the Nginx gateway; patient traffic reaches the gateway directly. The FastAPI backend and the local GPU avatar service share a private API network; only the backend has egress to model providers, and FHIR services are external to the backend. Daggers mark deployed components not exercised in this verification.

In Fig. 2, fixed questionnaire rules select the next enabled question and skip prefilled fields. Models classify prefill and extract chief complaints, indirectly affecting rule inputs, but do not select the next question. Keywords choose the questionnaire first; only no match permits evidence-grounded model routing. Failed chief assessment triggers handoff. Retrieval supports report drafts (Lewis et al. 2020), without establishing diagnoses or orders. We also prototyped an interview engine modeled on conversational diagnostic agents such as AMIE (Tu et al. 2025), in which a model selects each next question; it is selected through configuration behind the same interview interface, but it is still under development and was not evaluated.

2.2 Conservative EHR prefill normalization

UCC, launcher, and sandbox ingress deterministically number chronic-history items. The configured report-extraction adapter returns only indices and fixed categories: diabetes, chronic kidney disease, hyperlipidemia, cirrhosis, autoimmune disease, cancer, other chronic, exclude, or unclear. Listed categories become questionnaire options; other chronic retains original text. Any unclear item, provider failure, timeout, or invalid/incomplete indices leaves both history fields unfilled for patient answers. Otherwise, cancer/autoimmune/other chronic items also prefill `chronic_detail`; enabled but prefilled questions are skipped. No recognized condition leaves `chronic` unfilled, without inferring "none." Original history, excluded items, classifications, model/prompt provenance, and bounded usage remain available for physician review. Oversized inputs are explicitly rejected; failures never echo history. Normalization sends history to the report provider, defaults off, and was enabled for these runs.

2.3 Questionnaire and write-back contracts

The encoder maps fields to `linkId`s, options to codes, and conditions to `enableWhen`. Questionnaire defines the form; QuestionnaireResponse records answers (HL7 International 2019a, 2022). Source JSON drives execution. Questionnaire content is a replaceable input to FIRM: the chest-pain, headache, and abdominal-pain questionnaires used here were derived from a physician-authored source document, structured with LLM assistance, and extended by the development team; their clinical content was not evaluated in this study. Route selection pins the source-digest canonical version; write-back rejects drift and tags unpinned legacy resolution. R3 checks representation agreement.

After simulated clinician confirmation, opt-in structured write-back creates one transaction of the written-back resources in Table 1; Composition-only remains default. Conditional creation supports retries. Patient/encounter/institution/issuer checks precede persistence. Differential diagnoses, orders, and safety Flags are not created.

Absent-EHR placeholders are omitted. Out-of-grammar text uses coded "other" with nested original text, avoiding the validator's open-choice exception. Normalized answers retain `ehr-prefill` source and add normalization/prompt-version extensions. A runtime answer that FHIR would drop causes non-retryable rejection; unanswered branches do not block writing.

2.4 Synthetic inputs and model disclosure

Synthea generated the adult source cohort (Walonoski et al. 2018; The MITRE Corporation 2026). Conversion applies fictional Taiwanese identifiers/addresses and TW Core profiles (Ministry of Health and Welfare, Taiwan 2025), retaining seeds, mappings, dropped elements, and hashes. The 30 author-drafted chest-pain, headache, and abdominal-pain scenarios include expected stops and known gaps; the planned benchmark was unavailable. Separate Synthea modules supply consistent backgrounds without manually rewriting generated records. Converted bundles contain

Table 1 FHIR resources read for prefill (one `Patient/$everything` request) and written back, with assertion boundaries. *Absent from all synthetic inputs, so that read path was not exercised.

Resource	Content and boundary
Read for prefill	
Patient	Name, sex, and birth date.
Condition	Chronic history and route-specific cardiopulmonary, neurological, or abdominal history; current-encounter symptoms excluded.
MedicationRequest, MedicationStatement*	Current (active) and past (completed or stopped) medications.
AllergyIntolerance	Allergies.
Observation	Latest valid smoking status (LOINC 72166-2) mapped to the smoking options; unknown status is not prefilled. Blood group (LOINC 882-1/883-9)*.
Procedure*	Surgical history.
QuestionnaireResponse*	Prior history answers.
Written back	
QuestionnaireResponse	Ordered answers, source markers, and the pinned canonical reference.
Observation	Smoking status only for an explicit mapped patient selection; EHR text is not promoted to a new patient assertion.
Composition	Preliminary narrative, confirmed sections, and links to structured answers.
Provenance	Patient informant, AI assembly, and simulated clinician-review roles; no actual clinical approval.

Patient, Encounter, Condition, MedicationRequest, AllergyIntolerance, and Observation resources (Table 1). Input validation is separate from output validation.

Background audits are mechanical; approval requires two independent physicians reviewing current script hashes, bindings, and paths. 0 scripts qualify. Online Resource 1 provides prompts/settings. Commercial calls use Gemini's API (Google 2026); Gemma/Qwen use Ollama (Ollama 2026; Gemma Team et al. 2025; Qwen Team 2025). Requested/transmitted settings are distinguished: newer Gemini configurations omit temperature. Unrecorded settings are not inferred.

2.5 Requirements and acceptance criteria

Table 2 maps requirements to generated verdicts. E1–E5 are verification activities. E1 zero errors, E3 zero code changes, and E4's request-error stopping threshold were prespecified. Requirement-defined all-match/all-pass criteria for R2–R6, R7 behavioral agreement, and the investigator's R0 fixed-selection requirement were applied post hoc on 2026-10-14. These are software criteria, not prospective clinical thresholds; missing evidence is indeterminate.

Table 2 Requirements, criterion sources, and generated verdicts, grouped by the verification activity that supplies their evidence. R0 also uses E3 descriptive checks. R7 is reported separately for each replacement in Table 5.

ID	Requirement	Criterion source	Verdict
E1 — FHIR conformance and read-back: 200 templated sessions, 150 E2 sessions, and 10 smoking write-back sessions			
R1	TW Core structural/profile checks	Prespecified zero errors; no validator exceptions	Pass
R2	Write/read-back identity	Requirement-defined, applied post hoc	Pass
E2 — End-to-end sessions: 30 scripted scenarios × 5 repetitions with an LLM-simulated patient			
R0	Questionnaire rules select the next question	Principal investigator requirement; criteria applied post hoc	Pass
R3	Top-level path matches the FHIR Questionnaire	Requirement-defined, applied post hoc	Pass
R4	Extracted-fact traceability	Requirement-defined, applied post hoc	Pass
R5	Scripted safety-rule execution	Requirement-defined, applied post hoc	Fail
R6	End-to-end completion without handoff	Requirement-defined, applied post hoc	Pass
E3 — Model replacement: E2 scripts rerun with the alternate commercial and local open-weight configurations			
R7	Configuration-only substitution retaining default behaviour	Prespecified zero code lines; default-equivalence applied post hoc	Alternate: Pass Local: Fail
E4 — Local load replay of E2 turns: mock model at 10, 50, 100, 500 and real model at 1, 2, 4 concurrent sessions			
R8	National peak-load capacity	No national-demand criterion available	Not verifiable
E5 — MOHW exchange-hub integration			
R9	MOHW hub interoperability	Access prerequisite unavailable	Blocked

2.6 Verification activities

R0 audits saved calls for prefill normalization, chief extraction (at most once per session), report generation, or RAG translation, and requires identical repeated paths per script. Static inspection identifies the pure next-question selector. Prefill classification can change skip decisions; cross-model paths are descriptive. Bounded, incomplete usage cannot establish a complete call history.

E1 uses the pinned HL7/TW Core validator and generated definitions (HL7 International 2026; Ministry of Health and Welfare, Taiwan 2025). R1 requires zero written-resource errors/exceptions across templated, smoking write-back, and E2 cohorts. Because every templated and scenario record holds a coded smoking status that is prefilled, the separate smoking write-back cohort of 10 base-cohort patients outside the templated sample, with every smoking-status Observation removed, exercises the Observation write path. Checks use `-tx n/a`, without terminology resolution. R2 requires substantive write/read-back identity after excluding specified server metadata/identifiers. Missing outcomes cannot pass; supplemental E1 is counted once (Online Resource 1, Sections S3 and S7).

E2 spans invitation through read-back. A distinct Qwen model plays the patient from scenario facts, following simulated-patient evaluation (Tu et al. 2025; Johri

et al. 2025; Schmidgall et al. 2024). Invalid constrained answers receive explicit unknowns, never expected answers. R3 compares presented top-level questions with `enableWhen` paths derived from written QuestionnaireResponse answers, including EHR prefill. Recorded prefilled fields retain skip context for no-data items omitted from QR; nested `<linkId>.other` details are not questions. Runtime-answer and script-derived checks remain secondary. Expected chief stops are excluded; unexpected stops fail. R4 requires every persisted chief fact to have traceable utterance evidence. R5 requires all scripted red-flag and non-red-flag expectations, measuring rule execution rather than clinical sensitivity/specificity. R6 requires write-back and `completed` or `summary_ready`; expected stops and successful retries qualify, manual handoffs do not.

E3 retains executable fingerprints across model configurations. R7 requires zero code changes and default-equivalent R1–R6 verdicts and per-script completion, paths, and red flags; shared failures remain in the baseline. One-time adapter changes against `1b0eb1c` are documented separately.

E4 replays turns using fixed-latency mocks or real models on backend v0.19.0, one worker, stopping above 5% request errors. Session failure is distinct; this local rule does not define national capacity. E5 needs MOHW exchange-hub access. Frozen runs and source hashes generate results; repetitions are not independent clinical observations.

3 Results

3.1 R0: deterministic question selection

R0 is **Pass**. Among 1,460 saved calls, 0 tasks fall outside the allowed list; chief extraction peaks at 1 per session. 150 usage records are incomplete. Paths are identical in 30/30 scripts across 5 repetitions. E3 alternate Gemini matches default paths in 30/30; Gemma matches in 24/30. Routing records show 100 keyword selections, 0 extraction fallbacks, and 50 unknown sources; unknowns are not counted as keywords.

3.2 R1 and R2: written resources and read-back

R1 is **Pass** and R2 is **Pass**. Across 200 templated, 10 smoking write-back, and 150 E2 sessions, 1,090 written resources have 0 structural/profile errors and 0 validator exceptions under `-tx n/a` (Tables 3 and 4). Of 360 QuestionnaireResponses, 0 errors and 0 unverified responses are recorded. Warnings comprise 1,440 Provenance terminology and 360 Composition narrative warnings. The smoking write-back cohort wrote 10 smoking Observations with 0 errors. The pinned participant-type CodeSystem remains unavailable.

3.3 R3–R6: questionnaire, evidence, rules, and completion

R3 is **Pass**: written-FHIR paths match in 100/100 comparable sessions, versus 100/100 using runtime answer strings. 50 expected chief stops are excluded; 0 paths are unknown. Patient-entered controls agree in 45/45, whereas EHR-prefilled controls agree in 55/55 (Online Resource 1, Section S6.3). R4 is **Pass**, with traceability of

Table 3 E1 written-resource verification. Input checks are excluded; exceptions would prevent acceptance.

Cohort	Resource	n	With errors	Errors	Warnings
Templated cohort	Composition	200	0	0	200
Templated cohort	Provenance	200	0	0	800
Templated cohort	QuestionnaireResponse	200	0	0	0
Scenario cohort	Composition	150	0	0	150
Scenario cohort	Provenance	150	0	0	600
Scenario cohort	QuestionnaireResponse	150	0	0	0
Smoking writeback	Composition	10	0	0	10
Smoking writeback	Observation	10	0	0	0
Smoking writeback	Provenance	10	0	0	40
Smoking writeback	QuestionnaireResponse	10	0	0	0

Table 4 E1 workflow and semantic round trip. Read uses all sessions; write uses attempted writes; read-back uses returned references; round trip uses all captured submissions, including those without read-back.

Cohort	Sessions	Read	Write	Read-back	Round-trip
Templated cohort	200	100.0% (200/200)	100.0% (200/200)	100.0% (600/600)	100.0% (600/600)
Scenario cohort	150	100.0% (150/150)	100.0% (150/150)	100.0% (450/450)	100.0% (450/450)
Smoking writeback	10	100.0% (10/10)	100.0% (10/10)	100.0% (40/40)	100.0% (40/40)

100.0% (454/454). R5 is **Fail**; nonmatching scripts are AP-10, CP-09 (full denominators in Online Resource 1). R6 is **Pass**, with completion of 100.0% (150/150), including 50 expected chief stops. Simulator fidelity is reported separately.

3.4 Descriptive prefill evidence

Normalization supplies answers in 183/200 E1 and 90/150 E2 sessions. E2 has 0 unclear items and 0 failures; Gemma prefill has 1 timeout. Separate feasibility matching is 96.4% over 2308 provisionally labeled entries; truncated inputs include 15/105 guessed items (Online Resource 1).

3.5 R7: configuration-only replacement

R7 is **Pass** for alternate Gemini and **Fail** for local Gemma (Table 5). The comparison checks executable fingerprints and default-relative requirement/per-script outcomes. Relative agreement does not establish safety. Gemma differs on AP-03, AP-04, AP-07, CP-04, HA-02, HA-08. R3 is Pass/Pass for default/alternate configurations. Gemma's R3 is Fail (19/22), with 3 unexpected chief handoffs rather than coding divergences; completion is 27/30.

Table 5 E3 requirement verdicts by configuration. R7 compares each replacement with the default, including per-script completion, path, and red-flag outcomes; shared rule gaps do not themselves fail this relative check. Code lines exclude one-time adapter implementation.

Check	Default Gemini	Alternate Gemini	Local Gemma 3
R1	Pass	Pass	Pass
R2	Pass	Pass	Pass
R3	Pass	Pass	Fail
R4	Pass	Pass	Pass
R5	Fail	Fail	Fail
R6	Pass	Pass	Fail
Code lines	0	0	0
R7	Reference	Pass	Fail

3.6 R8 and R9: unavailable acceptance evidence

R8 is **Not verifiable** without national-demand evidence (`TODO_PEAK_DEMAND`); local stopping rules cannot substitute. R9 is **Blocked** without exchange-hub access. Local load, timing, and bounded usage/cost observations are descriptive (Online Resource 1).

4 Discussion

4.1 Principal findings

ELVINA keeps question selection under questionnaire rules and confines LLMs to bounded tasks: chief-complaint extraction with verbatim evidence, classification of EHR history into questionnaire options, and draft reports. The verification shows that this division, which FIRM prescribes, can be checked requirement by requirement. Question selection (R0) was Pass; written-resource conformance and read-back (R1, R2) were Pass and Pass; written and runtime question paths (R3) were Pass; fact traceability (R4) and end-to-end completion (R6) were Pass and Pass. Replacing the commercial model through configuration alone was Pass, supporting the claim that model-dependent components sit behind a replaceable adapter.

LLM-led diagnostic dialogue systems select their own questions (Tu et al. 2025), and evaluations that require models to elicit information through conversation report weaker performance than vignette-based testing (Johri et al. 2025; Schmidgall et al. 2024). ELVINA trades conversational flexibility for predictable, auditable question selection; that predictability is what made path-level verification possible. EHR-history classification does influence which history questions are skipped, but the next question is still chosen by the questionnaire rules. Whether fixed questionnaires elicit as much relevant history as free dialogue is a separate empirical question not addressed here.

4.2 Defects found by verification

The checks exposed defects that component tests had not revealed. First, the runtime evaluated free-text EHR history by substring while the written FHIR representation used coded answers, so an earlier verification round found written paths that disagreed with the questions actually asked. The remediation classifies EHR history entries into questionnaire options by index and leaves both history questions to the patient whenever any entry is unclear or classification fails; in the selected runs R3 is Pass against the unchanged written-resource criterion. Second, prefilled history was silently truncated at a fixed length; oversized input is now rejected explicitly. Third, comparing each scenario's FHIR record with its prefill showed that prefill ignored coded smoking-status Observations, so patients answered a question the EHR already held; prefill now maps the latest valid status and leaves unknown status to the patient. Fourth, chief-complaint findings for suspected pulmonary embolism and for abdominal pain radiating to the back with a known aneurysm were extracted correctly, but no rule combined them (R5, Fail: AP-10, CP-09). Candidate rules drafted from current guidance (Konstantinides et al. 2020; Wanhainen et al. 2024) await physician approval and new physician-written scenarios, and were not enabled. Finally, screening runs only at the chief complaint, so later questionnaire selections such as fainting do not raise an alert; simulated-patient answers that contradicted their scripts illustrated this gap.

Configuration-level replaceability did not guarantee behavioural equivalence. The local open-weight model was Fail for R7 (AP-03, AP-04, AP-07, CP-04, HA-02, HA-08), through chief-stage handoffs after timeouts, missed expected stops, and a conservative prefill fallback that returned chronic history to the patient; its R3 failures reflect those handoffs rather than path disagreement. Each model replacement therefore needs re-verification against the same requirements. The prototype modelled engine plugs into the same interview interface through configuration, which shows that the architecture also allows the interview engine to be replaced; because it was not evaluated, no behavioural claim is made for it. National capacity (R8) needs an authoritative demand target, and exchange-hub interoperability (R9) needs authorized test access.

4.3 Limitations and clinical review needs

Verdicts remain provisional pending physician review of the scenarios and independent audits. Requirement-defined criteria were applied post hoc, and clinician confirmation was simulated. The LLM-simulated patient sometimes contradicted its script; these answers were recorded as patient-reported and limit scenario fidelity rather than the path and write-back checks. EHR-history classification is not clinical confirmation: the default model assigned some entries to listed options that a provisional reference does not support, such as prediabetes as diabetes and colon polyps as cancer (67 entries in the templated cohort), excluded some chronic conditions, and was not fully consistent across sessions (Online Resource 1, Section S6). These answers are written back as unconfirmed EHR prefill, and the original EHR text remains available to clinicians; the classifier was not modified for this study. Its reference labels lack physician review, and synthetic disease-name distributions do not establish accuracy on real EHR data.

Structural/profile checks ran without terminology resolution. Synthetic demographics and sandbox interoperability do not establish clinical utility, production authorization, tenant isolation, browser SMART integration, or deployment readiness; terminology review, institutional governance, and human approval remain necessary.

5 Conclusion

ELVINA, built on the FIRM framework, keeps question selection under questionnaire rules and writes results back as TW Core FHIR resources, and it can be verified against explicit requirements. Requirements R0, R1, R2, R3, R4, R6 passed and R5 failed in this software verification; the failures identify concrete defects with documented remediations or pending clinical decisions. Replacing the commercial model by configuration was Pass, whereas a local open-weight model was Fail. Clinical validation, national-scale capacity (Not verifiable), and exchange-hub integration (Blocked) remain future work.

Supplementary information. Online Resource 1 documents methods, results, and limitations. Online Resource 2 supplies machine-readable synthetic materials, configurations, and provenance. Scenario approvals and the clinician screenshot remain pending.

Acknowledgements. Author confirmation is pending.

Declarations

Funding, competing interests, and contributions. Author statements and final approval are pending.

Ethics and consent. Only synthetic data were used. Authors must confirm the applicable institutional determination; no ethics approval is asserted.

Data availability. Online Resource 2 contains synthetic materials and selected results/provenance; full sessions and RAG source text are excluded. Medscape (Medscape 2026) redistribution rights require author confirmation (`TODO_AUTHOR`).

Code availability. Reproduction commands and source paths accompany Online Resource 2; the public repository location, archival identifier, and code/data licenses remain `TODO_AUTHOR`.

AI assistance. Claude Code and Codex assisted implementation, scenario drafting, orchestration, source checking, and manuscript revision. Model/tool disclosures and unrecorded versions appear in Online Resource 1, Section S5. Authors must confirm disclosure details, verify citations and content, and approve the text (`TODO_AUTHOR`); independent audits remain pending.

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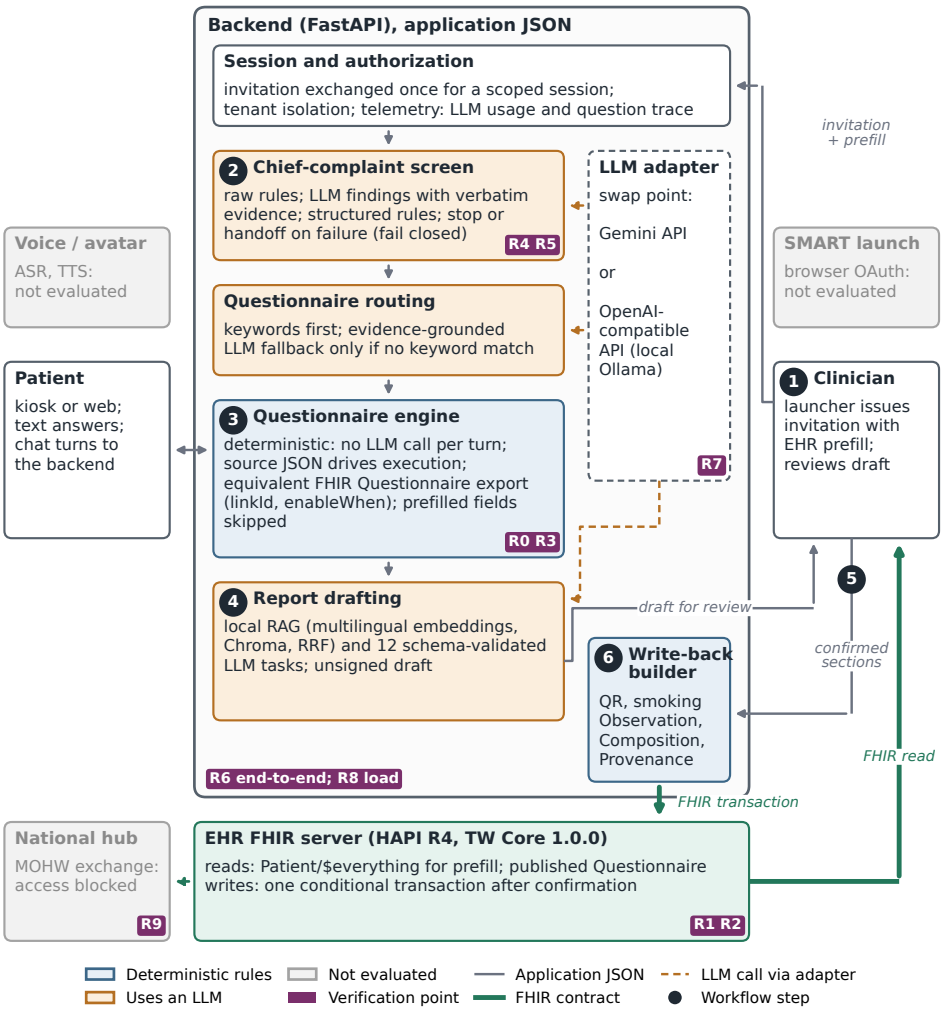


Fig. 2 FIRM workflow as implemented in ELVINA, with contracts and verification points. Workflow steps: (1) the clinician launcher reads EHR history and issues an invitation with prefill; (2) chief-complaint screening and questionnaire routing; (3) deterministic questionnaire turns; (4) report drafting; (5) clinician review and confirmation; (6) one FHIR write-back transaction. Source JSON drives the questionnaire engine and an equivalent FHIR Questionnaire is exported. Internal APIs use application JSON; FHIR defines the EHR boundary. Model-dependent components reach models only through the replaceable LLM adapter. Tags mark where requirements R0–R9 (Table 2) are checked; grey components were not evaluated.